

Exploring the metadata diversity in Social Sciences and Humanities catalogues: the case of GoTriple

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1 Introduction

Recent advances in technology have paved the way to a new context in the Cultural Heritage (CH) sector. Institutions have explored new ways to make their digital collections available so they can be reused by the research community (Mahey et al. 2019). Initiatives such as AI4LAM¹ and Collections as data promote the publication of digital collections suitable for computational use (Padilla et al. 2023). Other approaches are focused on the definition and establishment of best practices for ethical frameworks and data reuse such as the CARE principles (Carroll et al. 2020). They have opened new opportunities for researchers willing to study and analyse the content provided by the institutions and the outputs and data obtained as a result of research projects.

During the last years, there is growing interest in policy makers and society in Social Sciences and Humanities (SSH) research aiming at assessing the scientific, social, and political impact (Reale et al. 2017; Pedersen, Grønvad, and Hvidtfeldt 2020). Research in SSH promotes shared values including democracy, human rights, gender equality, and the richness of cultural diversity.²

Recent platforms facilitate the discovery and reuse of research artifacts focused on SSH. Some examples include GoTriple³ and the Social Sciences and Humanities Open Marketplace.⁴ In particular, GoTriple is a multilingual discovery platform providing access to publications, research data and project descriptions imported from aggregators, and semantically enriched and linked. Research data infrastructures have emerged in Europe during the last years. Some examples include the European data space for Cultural Heritage (DS4CH) and the European Open Science Cloud (EOSC).

1. <https://sites.google.com/view/ai4lam>

2. https://research-and-innovation.ec.europa.eu/research-area/social-sciences-and-humanities_en

3. <https://www.gotriple.eu/about>

4. <https://marketplace.sshopencloud.eu/>

This article intends to answer the following research questions: are existing discovery SSH platforms large enough to answer SSH research questions? What is the diversity of the content provided in SSH platforms? Are different languages sufficiently covered?

The main contributions of this article are: i) a framework to analyse the diversity of SSH catalogues; and ii) the data and code to reproduce the results. It intends to promote the reuse of this type of content for research purposes.

This article is organized as follows: after a description of the state of the art, we present the reproducible framework to analyse the diversity of catalogues and our results focused on GoTriple. The paper concludes with the main conclusions and final remarks.

2 Related work

This section describes an overview of SSH platforms and a literature review concerning the data publication an analysis focused on SSH content.

2.1 SSH platforms

This section provides an overview of SSH platforms, describing their main features. The examples are briefly described below.

Cross-domain approaches such as Wikidata enables the community to create and enhance metadata about resources such as datasets and tools (Candela et al. 2024; Pasquale et al. 2023). GoTriple enables users to discover and reuse open scholarly SSH resources, which are currently scattered across local and discipline-specific repositories, in multiple European languages. It makes use of innovative tools and services to support research and visualisation of search results. The Social Sciences and Humanities Open Marketplace provides a platform to enable users to publish tools, training materials, publications, datasets and workflows. It provides an API⁵ to access the data as well as a discovery interface enabling the filtering by fields such as language, activity or source, amongst others. Galaxy Community Hub is a platform to leverage analysis in SSH in various ways providing a flexible setup to create collaborative and runnable workflows (The Galaxy Community 2024).⁶ The Trans-Atlantic Platform for Social Sciences and Humanities (T-AP) is a collaboration between SSH and research funders from South America, North America, Europe and Africa offering joint research calls in areas of strong potential for international collaboration.⁷ Table 2.1 shows a summary of the selection of SSH platforms. Each column corresponds to a particular feature such as providing a multilingual content, providing access by means of an API, supporting collaborative edition functionalities or enabling the aggregation of content.

5. <https://marketplace.sshopencloud.eu/about/api-documentation>

6. <https://galaxyproject.org/community/sig/digital-humanities/>

7. <https://www.transatlanticplatform.com/>

Platform	Multilingual	API	Collaborative	Aggregator
Galaxy CommunitHub	No	Yes	Yes	No
GoTriple	Yes	Yes	No	Yes
SSH Open Marketplace	Yes	Yes	Yes	Yes
Trans-Atlantic Platform for SSH	No	No	No	No
Wikidata	Yes	Yes	Yes	No

Table 1: Overview of SSH platforms

2.2 Publication and analysis of SSH content

Institutions have started to publish FAIR (Findable, Accessible, Interoperable, Reusable) data (Wilkinson et al. 2016). Data in the CH have become crucial to promote research (Tasovac, Chambers, and Tóth-Czifra 2020; Candela et al. 2025). Previous work have analysed the publication and reuse of data in libraries (Gooding, Terras, and Ames 2025). Other approaches are focused on the enrichment and transformation of bibliographic information (Candela, Rosiński, and Margraf 2025). Rich and detailed documentation concerning how the content was curated and made available for the public has become crucial for potential researchers (Alkemade et al. 2025). In parallel, several initiatives have emerged to promote the reuse and enrichment of data such as the European Cloud for Heritage Open Science.⁸

Research in SSH involves the application of computational, linguistic and statistical techniques to extract meaningful insights from various types of data such as metadata, data and text. Previous work have tackled the analysis of metadata and content using diversity indices (Colwell, Coddington, and Hawksworth 1994). They have been traditionally used in biodiversity as an estimation of the species present in an ecosystem. Richness is defined as the number of species present weighting all groups equally, regardless their relative abundances. In contrast, diversity indices measure richness and evenness, giving more relevance to abundant species (Hill 1973). Diversity indices were employed to measure diversity of digital collections in libraries including textual content and metadata (Carrasco, Candela, and Marco-Such 2025).

Other initiatives have explored search and retrieve techniques across diverse information platforms, such as digital humanities archives, and digital libraries, synthesizing the resources discovered (Momeni and Hoeber 2025). Additional initiatives have assessed the quality of metadata by applying a data quality criteria including a wide diversity of metrics such as completeness and enrichment with external repositories (Candela 2023; Färber et al. 2018; Dişli, Candela, et al. 2025).

Despite all this work, to our best knowledge none of them provide a systematic, reproducible and computational approach to analyse SSH metadata. This work can be useful for other institutions willing to publish their catalogues as well as the research community to identify new ways to analyse data.

8. <https://www.echoes-ecch.eu/>

3 A framework to analyse the diversity of SSH metadata

The methodology proposed in this article to analyse the diversity of metadata provided by SSH catalogues works in three steps: i) identification of topics; ii) retrieve and prepare data; and iii) analyse data and report. The steps are described below.

3.1 Identification of topics

The first step corresponds to the identification of SSH topics. Previous work have analysed and classified research activities in SSH (Bakhshi, Breckon, and Puttick 2021; Münster and Terras 2019). Some examples include Economy, Education, Media and Communications, History, Ethics or Literature.

A selection of topics can be retrieved using existing databases or public encyclopedias.⁹ Other sources can be previous research work focused on the identification of research topics in SSH. Moreover, cross-domain repositories such as Wikidata also provides a selection of topics created by the community.¹⁰

As an example, Appendix 5 includes an overview of research SSH topics.

3.2 Retrieve and prepare data

The second steps corresponds to the retrieval of the data according to the topics selected in the previous step. The data can be retrieved using an API. There are different APIs that can be used to retrieve the metadata such as OAI-PMH and SPARQL (Dişli, Osti, et al. 2025). The API receives a set of parameters as input such as the term query, number of results or language. A common and user-friendly output format used by the projects is JSON, however, other type of formats can include XML and Resource Description Framework (RDF) (World Wide Web Consortium 2014).

The use of APIs sometimes may be restricted in terms of number of requests during a period of time (e.g., per day), subjects or maximum number of results (e.g., 1,000). Since this can be a problem for the researchers when analysing the topics, in some cases, contacting the staff of the project may solve this problem. Appendix B describe the queries used to query the GoTriple discovery platform using the research topics included in Table 5.

Each of the records retrieved may include a selection of metadata fields to describe it. Depending on the purpose of the analysis, only part of them might be useful. For diversity purposes, fields such as keywords, subjects and language can be used. In some cases, aggregators may create additional metadata fields to enrich the content.

9. See, for example, <https://libguides.umflint.edu/topics> and <https://www.gotriple.eu/disciplines>

10. <https://www.wikidata.org/wiki/Q33122512>

In order to reuse the data, a preprocess and cleaning task is required in order to handle missing values, removing duplicates, fixing errors, normalizing data to improve data quality and ensure reliable, accurate results.

3.3 Analyse data and report

Once the data is extracted, the next step involves the analysis of the data. First of all, a set of criteria to analyse the metadata.

Diversity indices has been traditionally used to analyse the diversity of species in a ecosystem. This can be adapted to SSH by analysing a selection of metadata fields such as language and keywords. While there are different indices, the Shannon index has been previously used for several purposes such as measure lexical and metadata diversity in digital collections (Carrasco, Candela, and Marco-Such 2025; Jarvis 2013).

Shannon diversity index is a key ecological metric measuring biodiversity, combining species richness (or number of species) and species evenness (abundance distribution) into a single value, where higher figures represent greater diversity (Shannon 1948). Let p_n be the proportion of each species in the community, the Shannon diversity index is computed using the following formula:

$$H = - \sum_{n=1}^N p_n \log p_n \quad (1)$$

Diversity indices can be useful to avoid silences and biases in data, as recommended by best practices and guidelines (Padilla et al. 2023; Carroll et al. 2020). This is interesting for the research community as a way to provide a transparent method to measure diversity but also to create awareness of these type of data issues.

Finally, the last task corresponds to the reporting of the analysis by means of textual documentation. The results can be made available in research platforms such as Zenodo.

4 Results

The methodology proposed in Section 3 was applied to the GoTriple discovery platform. As an example of how to apply the framework proposed in this work, we selected 4 examples as shown in Table 2, covering different subjects and number of results. The values in column query were used to retrieve the data from the platform. A set of JSON files were retrieved with the help of the GoTriple admin staff including the results.¹¹

Then, after an analysis of the metadata fields provided by each of the results, a selection of fields were extracted to apply the diversity index: i) keywords; and language. For the analysis a period of time were established between 1998 and 2020. Since GoTriple is a multilingual repository, keywords are provided

11. Note that the data was extracted in October 2025.

Example	Subject Area	Subject	Query	N. of results
1	Society, Culture, and Identity	Race, ethnicity, and decolonization	"decolonization" OR "postcolonial studies" OR "race and ethnicity" OR "decolonial theory"	6861
2	Digital Transformation and Technology	Digital humanities	"digital humanities" OR "text mining" OR "data visualization" OR "digital archives"	12325
3	Interdisciplinary and Emerging Areas	Health humanities	"health humanities" OR "medicine and literature" OR "bioethics"	8637
4	Interdisciplinary and Emerging Areas	Indigenous knowledge systems	"indigenous knowledge" OR "traditional ecological knowledge" OR "epistemic justice"	5142

Table 2: Queries and results employed to retrieve the data from GoTriple

in several languages. As a result, we focused our analysis on particular languages (e.g., Spanish and English) to measure the diversity of keywords. The information about the language for keywords is not provided so we use an open software library to detect the language selected for the analysis. Table 3 shows the results obtained for the keywords metadata field. Figure 1 shows a selection of plots showing how the diversity and richness change over time.

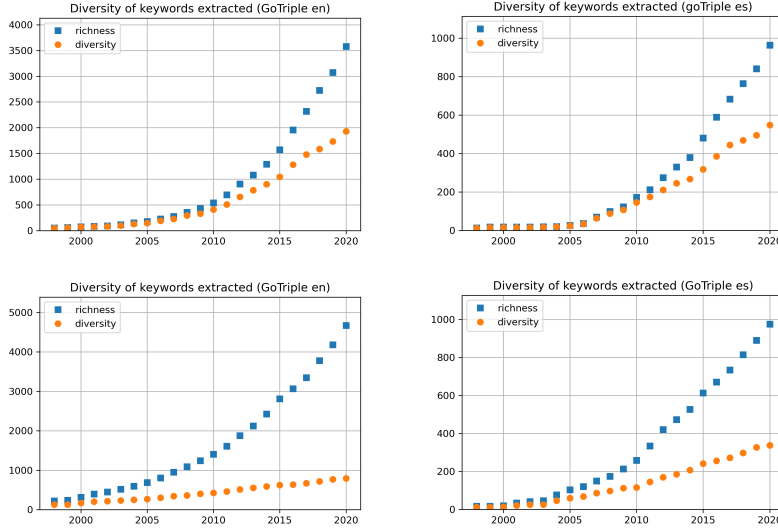


Figure 1: Left: Richness and Shannon diversity for the examples 1 and 3, using as input the field keywords in English. Right: Richness and Shannon diversity for the examples 1 and 3, using as input the field keywords in Spanish.

Table 4 shows the results after the application of the framework to the metadata field language. Figure 2 shows the plots for the language metadata and diversity analysis.

The results obtained showed that in the case of keywords are diverse. The smaller ratio between the Shannon diversity and richness for the example 3 (about 17-35%) in comparison to the ratio for the example 4 (50-65%) might be caused to different reasons such as the curation of the content and the source of the records. In the case of the languages used in the records of GoTriple

Example	Language	Richness	Shannon	D/R
1	English	3,563	1,929	0.54
1	Spanish	944	527	0.56
2	English	5,191	2,186	0.42
2	Spanish	776	504	0.64
3	English	4581	782	0.17
3	Spanish	972	339	0.35
4	English	2,889	1,440	0.5
4	Spanish	404	264	0.65

Table 3: Metadata diversity figures obtained as a result of the analysis of the examples included in Table 2 for the metadata field keywords. Note that the table show results for English and Spanish. The last column corresponds to the rate the Shannon diversity index and the Richness value.

Example	Richness	Shannon	D/R
1	25	7.47	0.30
2	23	8	0.34
3	22	7.37	0.34
4	21	6.76	0.32

Table 4: Metadata diversity figures obtained as a result of the analysis of the examples included in Table 2 for the metadata field language. The last column corresponds to the rate the Shannon diversity index and the Richness value.

they follow a similar pattern for all the examples. There is a group of languages which is more representative in the whole set of records.

A GitHub repository was created including all the code and data to reproduce the results.¹²

4.1 Discussion

This section describes the limitations of this work in terms of scope, infrastructure and results obtained.

The GoTriple infrastructure has some limitations that are described below.

- The API only retrieves 10,000 records per query.
- The data is handled by gotriple services and some additional data is automatically created such as language or topics.
- GoTriple provides different types of records. In this work, all the results are articles. This can be done when using the API.

12. <https://github.com/hibernator11/fasca-diversity-gotriple>

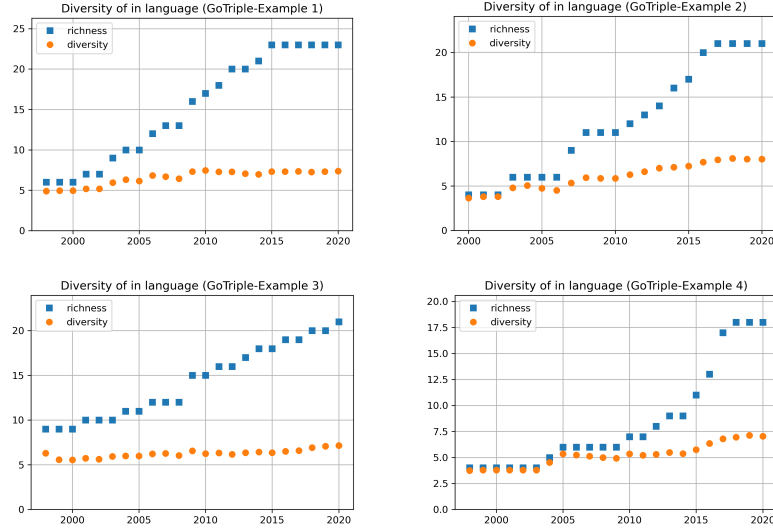


Figure 2: Richness and Shannon diversity for the examples selected.

Concerning the cleaning and preprocessing tasks, we identified several issues that are described below. In order to identify the language of the keywords, we used a Python libraries that in some cases provided false positives when using Spanish.¹³ Additional work is required in order to finetune the identification of the language such as using Large Language Models (LLM). In some cases the year column was empty. Since we used a plot to show how diversity and richness change over time, we removed the rows with an empty value in the year column.

The methodology proposed in this work can be adapted to other repositories and domains by refining each of the steps provided.

5 Conclusions

SSH discovery platforms aggregate results such as articles, data and documentation from the research community. Recent initiatives promote the reuse and analysis of the data available in order to create prototypes, create awareness and increase the impact. This work introduced a framework to measure the diversity of metadata in SSH catalogues. Diversity indices has been extensively used employing different type of content as input such as text and metadata. Diversity is essential in the current context to avoid biases in data. We used GoTriple as data source for our analysis. Results showed that....

Future work to be explored include the application of the framework to other platforms, the inclusion of additional data quality criteria to assess the metadata and the provision of the results using a machine-readable format.

13. <https://pypi.org/project/langdetect/>

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A Research topics in SSH

Table 5 shows an overview of research topics in SSH.

B Queries

Table 6 describe the queries used to answer the research topics included in Table 5 using as data source the discovevy platform GoTriple.

Category	Topics
Society, Culture, and Identity	Gender and sexuality studies - including intersectionality, queer theory, and masculinity studies Race, ethnicity, and decolonization — decolonial theory, postcolonial studies, and global south perspectives Migration and diaspora studies - transnationalism, refugees, and borders Cultural heritage and memory studies - digital archives, post-conflict memory, and cultural restitution
Global Challenges and Social Change	Climate change and sustainability - environmental humanities, climate justice, and eco-criticism Inequality and social justice - economic disparities, poverty, and social policy Urbanization and smart cities - spatial justice, housing, and the future of work Global governance and democracy - populism, authoritarianism, and human rights
Digital Transformation and Technology	Digital humanities - computational text analysis, data visualization, and digital archives AI and society - ethics of artificial intelligence, algorithmic bias, and human-machine interaction Media and communication studies - social media, misinformation, and digital publics Cyberculture and virtual communities - gaming, online identity, and digital creativity
Economy, Work, and Policy	Political economy - globalization, neoliberalism, and financial crises Future of work - automation, platform economies, and labor precarity Public policy and governance - evidence-based policy, behavioral economics, and institutional trust Education and learning - digital pedagogy, access and equity, and critical education studies
History, Philosophy, and Thought	Global and transnational history - empire, trade, and knowledge exchange Philosophy of technology and ethics - posthumanism, AI ethics, and moral responsibility Critical theory and epistemology - power, discourse, and the politics of knowledge History of science and medicine - pandemics, scientific authority, and biopolitics
Interdisciplinary and Emerging Areas	Health humanities - intersections between medicine, literature, and ethics Environmental social sciences - human-nature relationships, resilience, and adaptation Posthuman and anthropocene studies - rethinking the human in planetary contexts Indigenous knowledge systems - traditional ecological knowledge and epistemic justice

Table 5: Overview of research topics in SSH classified by categories

Subject Area	Subject	Query	N. of results
Society, Culture, and Identity	Gender and sexuality studies	"gender studies" OR "queer theory" OR "sexuality" OR "intersectionality"	35400
Society, Culture, and Identity	Race, ethnicity, and decolonization	"decolonization" OR "postcolonial studies" OR "race and ethnicity" OR "decolonial theory"	6861
Society, Culture, and Identity	Migration and diaspora studies	"migration" OR "diaspora" OR "refugees" OR "transnationalism"	128611
Society, Culture, and Identity	Cultural heritage and memory studies	"cultural heritage" OR "collective memory" OR "heritage restitution" OR "memory politics"	41910
Global Challenges and Social Change	Climate change and sustainability	"climate change" OR "sustainability" OR "climate justice" OR "environmental humanities"	315293
Global Challenges and Social Change	Inequality and social justice	"inequality" OR "social justice" OR "poverty" OR "social policy"	153845
Global Challenges and Social Change	Urbanization and smart cities	"urbanization" OR "smart cities" OR "urban justice" OR "housing policy"	42652
Global Challenges and Social Change	Global governance and democracy	"democracy" OR "global governance" OR "authoritarianism" OR "human rights"	154745
Digital Transformation and Technology	Digital humanities	"digital humanities" OR "text mining" OR "data visualization" OR "digital archives"	12325
Digital Transformation and Technology	AI and society	"artificial intelligence" OR "AI ethics" OR "algorithmic bias" OR "human-machine interaction"	45111
Digital Transformation and Technology	Media and communication studies	"social media" OR "misinformation" OR "digital communication" OR "public sphere"	86408
Digital Transformation and Technology	Cyberculture and virtual communities	"cyberculture" OR "virtual communities" OR "online identity" OR "digital creativity"	2397
Economy, Work, and Policy	Political economy	"political economy" OR "globalization" OR "neoliberalism" OR "financial crisis"	109738
Economy, Work, and Policy	Future of work	"automation" OR "future of work" OR "platform economy" OR "gig work"	17294
Economy, Work, and Policy	Public policy and governance	"public policy" OR "governance" OR "institutional trust" OR "behavioral economics"	167470
Economy, Work, and Policy	Education and learning	"digital education" OR "critical pedagogy" OR "educational equity"	6383
History, Philosophy, and Thought	Global and transnational history	"global history" OR "transnational history" OR "empire" OR "knowledge exchange"	44304
History, Philosophy, and Thought	Philosophy of technology and ethics	"philosophy of technology" OR "AI ethics" OR "posthumanism" OR "moral responsibility"	4760
History, Philosophy, and Thought	Critical theory and epistemology	"critical theory" OR "power and discourse" OR "knowledge politics" OR "epistemology"	28773
History, Philosophy, and Thought	History of science and medicine	"history of science" OR "biopolitics" OR "pandemics" OR "medical humanities"	18577
Interdisciplinary and Emerging Areas	Health humanities	"health humanities" OR "medicine and literature" OR "bioethics"	8637
Interdisciplinary and Emerging Areas	Environmental social sciences	"environmental sociology" OR "human-nature relations" OR "resilience"	66867
Interdisciplinary and Emerging Areas	Posthuman and Anthropocene studies	"posthumanism" OR "anthropocene" OR "planetary ethics"	7477
Interdisciplinary and Emerging Areas	Indigenous knowledge systems	"indigenous knowledge" OR "traditional ecological knowledge" OR "epistemic justice"	5142

Table 6: Queries and results provided by GoTriple according to the research topics